Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

B.Tech BCSE497J – Project-I

NeuroScan AI: AI-Powered Brain Tumour Detection, Classification and Segmentation Platform

Team members

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Faculty guide

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Based on two research manuscripts: a leakage-controlled benchmark of 2D backbones for detection and subtype classification, and a region-adaptive CNN-transformer ensemble for 3D segmentation with external validation

Approval Mail From Guide

The Fall Semester Project Review 3 for Project ID <<25000UG
approved.

- **Meeting with Guide Frequently:** Satisfied / Not Satisfied
- **PowerPoint Presentation:** Approved / Not Approved
- **Project Demonstration:** Approved / Not Approved
- **Final Project Report:** Satisfied / Not Satisfied
- **Project Outcome:** Ongoing / Completed

All the above aspects are addressed and updated by the project team.

Aim

To build and rigorously evaluate an AI system that detects brain tumours in MRI, classifies them as glioma, meningioma or pituitary tumour, and segments the whole tumour, tumour core and enhancing tumour in 3D, selecting every deployed model through a controlled, leakage-aware benchmark.

453,623

primary brain and other CNS tumours diagnosed in the United States, 2016 to 2020 [1]

26.3%

of these tumours are gliomas [1]

About 80%

of malignant brain tumours worldwide are gliomas [2]

1. Detect and classify in 2D

Benchmark nine ImageNet-pretrained CNN, transformer and hybrid backbones for tumour detection and glioma, meningioma or pituitary classification on deduplicated public MRI slices (Paper 1).

2. Segment in 3D

Fuse two 3D CNNs and a transformer per tumour region for whole tumour, tumour core and enhancing tumour, and test them on BraTS 2020 and on external BraTS-Africa cases (Paper 2).

3. Deploy

Serve the most efficient models, EfficientNetB0 and SegResNet, in a secure web application with a linked multi-planar and 3D viewer and PDF and CSV reports.

Motivation. MRI is the main non-invasive modality for characterising brain tumours [3], which are classified by cell of origin and molecular features [4]. Manual reading is slow and varies between and within readers [5], and reliable automation can support faster, more consistent assessment [6, 7].

Abstract

Problem. Published brain tumour MRI results cannot be compared directly: studies differ in data, splits, training budgets and metrics, and near-duplicate images leak between training and test partitions [8, 9].

Paper 1: 2D detection and subtype classification

- Nine backbones (six CNNs, two transformers, one hybrid), one code path and an equal number of optimiser steps
- Label-aware perceptual-hash deduplication: 3,253 to 1,185 detection images and 5,828 to 4,730 subtype images; group-aware five-fold split with a held-out test set
- Fold ensembles, bootstrap 95% CIs, exact McNemar tests with Holm correction, calibration, Grad-CAM and background-only shortcut audits

99.42% detection accuracy of CoAtNet-0, EfficientNetB0 and ViT-B/16 on 171 test images (95% CI 98.25 to 100%); no pair of models differs significantly; background-only models beat the majority-class rate for 7 of 9 detection and 9 of 9 subtype backbones

Paper 2: 3D segmentation with external validation

- nnU-Net and SegResNet (CNNs) and Swin UNETR-v2 (transformer) at native 1 mm resolution [10–12]
- Region-adaptive ensemble: one weight vector and threshold per region, chosen on 56 validation cases
- 56 internal BraTS 2020 test cases and 60 external BraTS-Africa cases [13]; member disagreement tested as a quality-control signal

0.993 AUC of member disagreement for flagging failed cases internally (0.870 externally); the ensemble (Dice 0.8530) beats two members but not SegResNet alone (0.8717)

Deployment. EfficientNetB0 (detection and classification) and SegResNet (segmentation) run in a FastAPI and React application; 64 backend and 39 frontend automated tests pass.

Literature Review

Study	Task and data	Method	Reported result	Limitation addressed here
Zacharaki et al., 2009 [14]	Glioma vs metastasis, glioma grade; 102 tumours	Shape, intensity and texture features; SVM with feature elimination	85% and 88% (leave-one-out)	Hand-crafted features transfer poorly [15, 16]
Cheng et al., 2015 [17]	Three-class subtype; 3,064 T1CE slices, 233 patients	Augmented tumour ROI, ring partition; intensity, GLCM, bag-of-words + SVM	Up to 91.28%	Needs a tumour ROI; features redesigned per task
Seetha and Raja, 2018 [18]	Tumour vs non-tumour; Radiopaedia, BRATS 2015	ImageNet-pretrained CNN, last layer trained	97.5% (training)	No held-out estimate
Talo et al., 2019 [19]	Normal vs abnormal; 613 MR images	ResNet34 transfer learning [20, 21]	100% (five-fold)	One small collection
Deepak and Ameer, 2019 [22]; Ghassemi et al., 2020 [23]	Three-class subtype; figshare 3,064 slices	GoogLeNet features + classifiers; CNN pretrained as a GAN discriminator	98% mean (Deepak)	One architecture each
Rehman et al., 2020 [24]; Noreen et al., 2020 [25]	Three-class subtype; figshare 3,064 slices	Fine-tuned VGG16 [26]; Inception-v3 [27] and DenseNet201 [28] features	98.69%; 99.34% and 99.51%	No deduplication, CIs or tests
Khan et al., 2020 [29]	Tumour classification; BraTS 2015, 2017, 2018	VGG16 and VGG19 features, ELM-based selection	97.8%, 96.9%, 92.5%	Multi-stage tuned pipeline
Tummala et al., 2022 [30]; Asiri et al., 2024 [31]	Three-class (3,064); four-class (2,870)	ViT ensemble [32]; Swin Transformer [33]	98.7%; 97%	No CNN baseline under one protocol
3D CNNs [10, 11, 34, 35]	3D medical segmentation	U-Net, 3D U-Net, SegResNet, self-configuring nnU-Net	Strong CNN baselines	Mostly tested on internal splits
3D transformers [12, 36–39]	3D brain tumour segmentation	UNETR, Swin UNETR (v2), TransBTS, TransUNet	Long-range context	Fused with CNNs by plain averaging [40]
Kapoor and Narayanan, 2023 [8]; Fateh et al., 2026 [41]	Reliability of ML results; BRISC curation	Leakage audit across 17 fields; duplicates removed before splitting	Leakage inflates scores	No architecture benchmark

Research Gap

	Gap in the literature	How this project addresses it
1	No controlled comparison. Most studies test two or three models with their own data, splits and budgets [42, 43].	Nine 2D backbones and three 3D networks, each set under one code path and identical splits
2	Data leakage. Combined public collections hold exact and near-duplicate images that are rarely removed before splitting [8, 9, 41].	Label-aware perceptual-hash deduplication, similarity groups, group-aware split and a leakage audit
3	Single-run leaderboards. Results are quoted from one run, without confidence intervals, tests, per-class metrics or calibration.	Five-fold CV, fold ensembles, bootstrap CIs, McNemar and Wilcoxon tests with Holm correction, ECE, per-grade results
4	Shortcuts unchecked. Models can exploit borders, hospital markers or scanner cues instead of pathology [44–46].	Grad-CAM, occlusion, head and background masking, and background-only retraining
5	No external check in 3D. 3D results are mostly internal, models degrade on new populations and scanners [45], and per-model uncertainty is a weak error signal [47].	External BraTS-Africa test set, per-region fusion, and member disagreement as a quality check
6	No path to use. Benchmarked models are rarely turned into tools that accept MRI volumes and protect patient data.	A secure web application with a multi-planar and 3D viewer and reports

Objectives

1. Benchmark nine 2D backbones for tumour detection and subtype classification with identical splits and an equal number of optimiser steps.

2. Remove leakage before splitting through label-aware perceptual-hash deduplication, similarity grouping and a group-aware split, verified by a leakage audit.

3. Evaluate with five-fold cross-validation, fold ensembles, bootstrap CIs, Holm-corrected McNemar tests, calibration, efficiency and shortcut audits.

4. Fuse two 3D CNNs and a transformer per tumour region on BraTS 2020, and test every model internally, externally on BraTS-Africa, for failure detection and for cost.

5. Deploy the most efficient models in a secure, tested web application with a multi-planar and 3D viewer and reports.

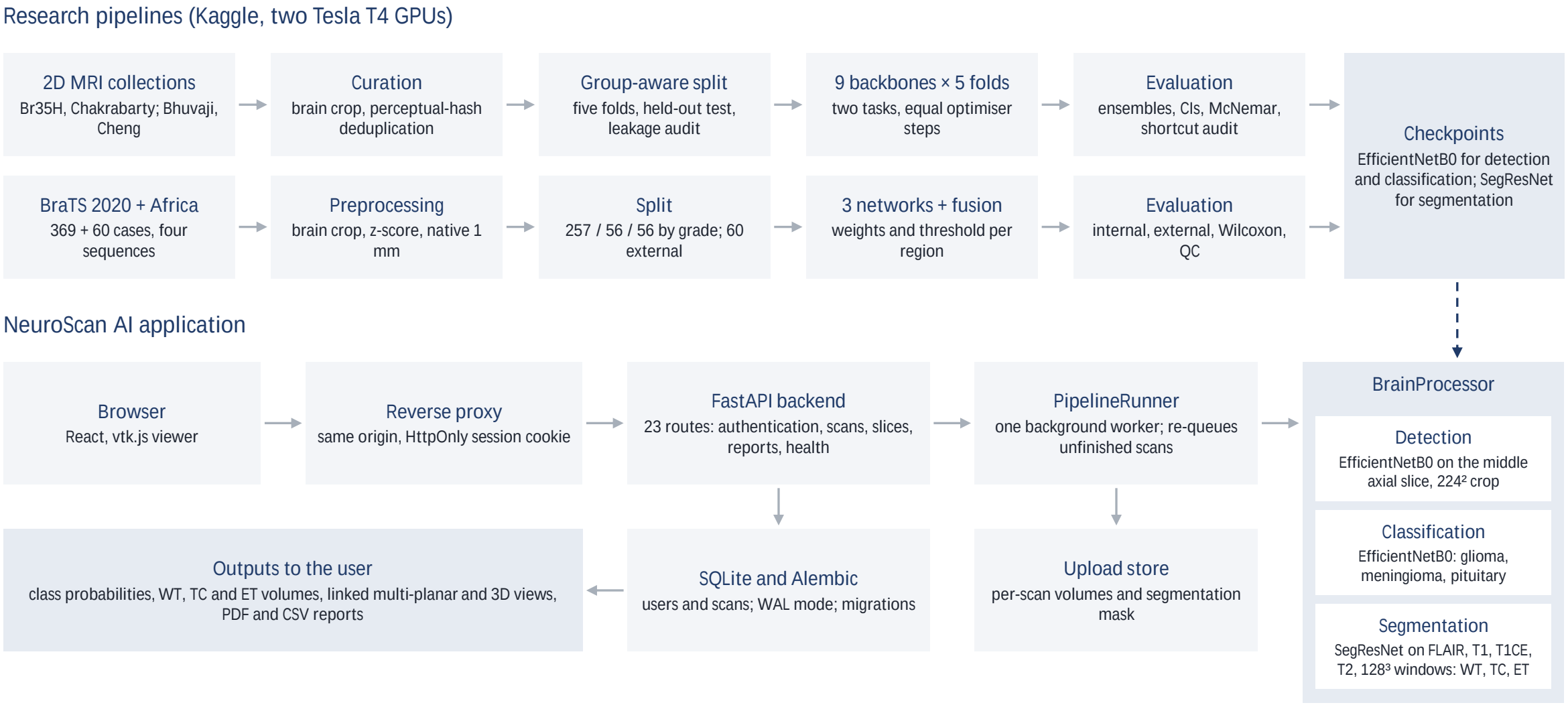
Sustainable Development Goals and Technology Readiness Level





- SDG 3: Good Health and Well-being; SDG 9: Industry, Innovation and Infrastructure
- TRL 4: technology validated in the laboratory
- Outcomes: two journal manuscripts (Scopus) prepared for *Engineering Applications of Artificial Intelligence* (Elsevier); an open-source web application

Framework/Architecture/Block Diagram



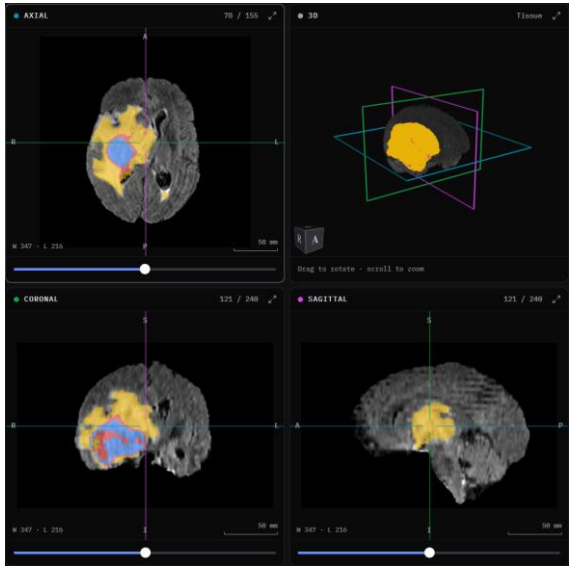
Functional Requirements and Modules

Functional requirements

- Register, sign in and out, change the password; optional TOTP two-factor authentication
- Upload one to four NIfTI volumes (FLAIR, T1, T1CE, T2) per case; modality detected from the file name
- Process each scan in the background with live status, step and progress; retry on failure
- Detect a tumour, classify its type with class probabilities, and segment WT, TC and ET with volumes
- View linked axial, coronal, sagittal and 3D panes with overlay, contrast, zoom and ruler
- Search, rename and delete scans; dashboard with counts, recent scans and a volume chart
- PDF and CSV report per scan; CSV export of all scans
- Operator account recovery from the command line

Modules

- Data curation: brain crop, perceptual-hash deduplication, label-noise exclusion, group split
- 2D benchmark: 9 backbones × 5 folds plus a background-only run; 54 runs per task
- 3D study: three networks, per-region fusion, internal and external tests, disagreement QC
- Inference engine: BrainProcessor with safe weight loading, the training crop and the mask in FLAIR space
- Backend and pipeline: FastAPI (23 routes), one worker, SQLite in WAL mode, Alembic
- Security: bcrypt, revocable JWT cookies, CSRF header, lockout, rate limits, TOTP
- Viewer: React and vtk.js; linked multi-planar and 3D views, slab modes, ruler
- Reports: PDF MRI report with key images; CSV export with formula escaping



NeuroScan AI viewer: linked axial, 3D, coronal and sagittal panes with the WT, TC and ET overlay on a BraTS case

Data Curation and Leakage Control

1. Brain crop
greyscale, 5×5 Gaussian, threshold 45, largest contour [48]

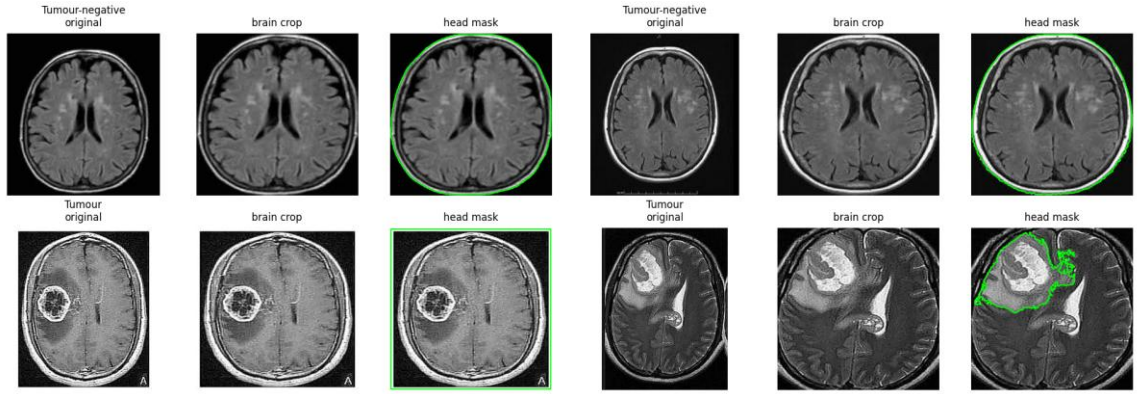
2. Perceptual hash
64-bit pHash [49] and a 64×64 thumbnail per image

3. Deduplicate
same label, Hamming ≤ 5 [50]: keep one; cross-label NCC ≥ 0.98 [51]: drop both

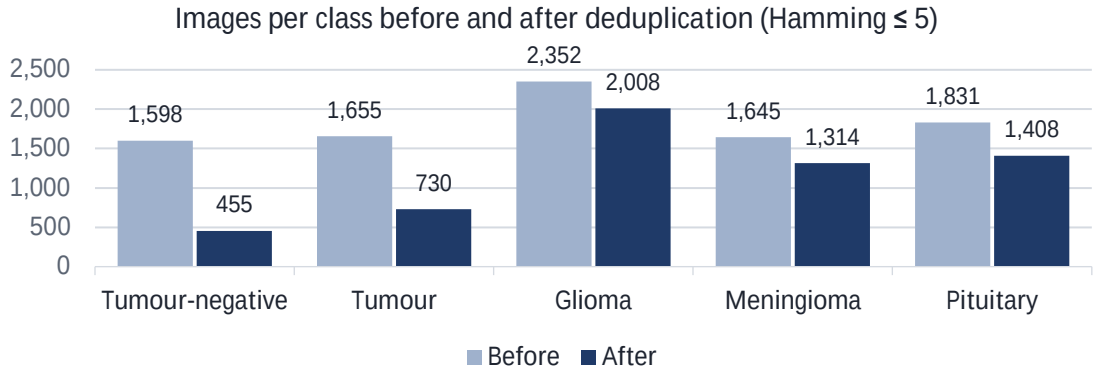
4. Group
Hamming ≤ 8 or NCC ≥ 0.98: 821 and 1,867 groups as a patient proxy

5. Split and audit
stratified group k-fold [52, 53]: 15% test, five folds; no test image near development

Task	Sources	Images	After deduplication	Dev.	Test
Detection	Br35H [54], Chakrabarty [55]	3,000 + 253	1,185 (−63.57%)	1,014	171
Subtype	Bhuvaji [56], Cheng [17, 57, 58]	2,764 + 3,064	4,730 (−18.84%)	4,685	45



Original image, brain crop used for training, and head mask used by the attention audit



- Eight of 31 cross-label pairs in the subtype pool were label noise; the detection pool had none.
- One subtype group held 1,431 images, so the subtype test set has only 45 images (1 glioma, 13 meningioma, 31 pituitary).
- The all-pairs search grows with the square of the pool; multi-index hashing [59] would scale it to larger collections.

Backbones, Training and Evaluation

Backbone	Family	Params (M)	GFLOPs	Latency (ms)	Images/s
ResNet50 [20]	CNN	23.51	8.17	9.57	777
VGG16 [26]	CNN	134.26	30.93	7.72	387
DenseNet121 [28]	CNN	6.95	5.67	23.58	657
EfficientNetB0 [60]	CNN	4.01	0.77	13.13	1,393
MobileNetV3-Large [61, 62]	CNN	4.20	0.43	10.55	2,301
InceptionV3 [27]	CNN	21.79	11.42	17.63	512
ViT-B/16 [32, 63]	Transformer	85.80	22.54	10.95	272
Swin-Base [33]	Transformer	86.74	30.86	36.29	198
CoAtNet-0 [64]	Hybrid	26.67	8.42	16.53	513

CNNs stack learned convolutional filters [65, 66]; transformers relate all image patches through self-attention. All models are ImageNet-pretrained [67, 68] torchvision or timm models in PyTorch [69], with a one-output (detection) or three-output (subtype) head.

Cost was measured on one Tesla T4 in half precision: latency at batch size 1, throughput at batch size 32. It spans a factor of about 72 (0.43 to 30.93 GFLOPs) while detection accuracy spans 2.34 points, and latency does not follow size: VGG16 (134 M parameters) takes 7.7 ms, DenseNet121 (6.95 M) 23.6 ms.

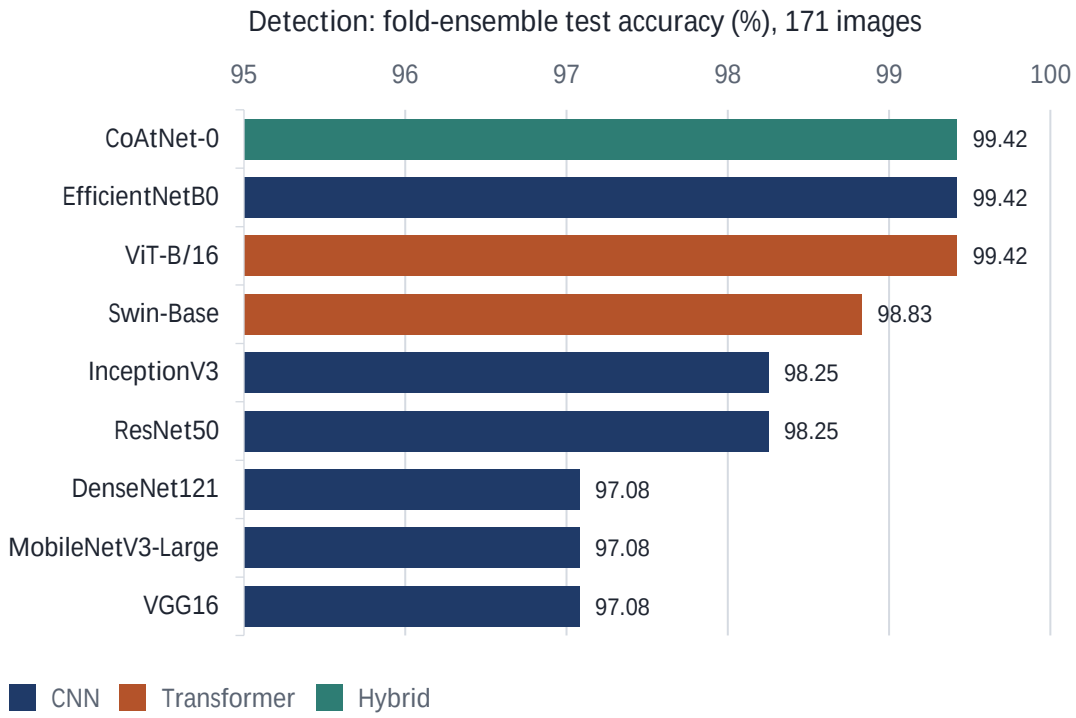
Identical training: 54 runs per task

- AdamW [70], learning rate 3×10^{-4}, weight decay 0.01
- 20 epochs; effective batch 64 by gradient accumulation, so every model takes the same optimiser steps
- One-epoch warm-up and cosine decay [71]; half precision [72]; gradient clipping at 1.0 [73]
- Flip, ±10° rotation, ±20% brightness and contrast; checkpoint at the lowest validation log-loss [74]

Evaluation protocol

- Temperature scaling [75] and Youden threshold [76] fitted on each validation fold
- Five fold models averaged into one fold-ensemble prediction per test image
- Accuracy, macro F1, AUC [77, 78], MCC [79, 80] and ECE with 15 bins [81]
- Bootstrap 95% CIs from 2,000 resamples [82]; exact McNemar tests [83], Holm-corrected over 36 pairs [84]

Detection and Subtype Results



- CoAtNet-0 missed one tumour; EfficientNetB0 and ViT-B/16 detected all 107 tumours and raised one false alarm.
- Every AUC lies between 0.9974 and 0.9997; the fold-to-fold standard deviation of accuracy, 0.32 to 2.20 points, matches the gaps between backbones.
- On the test images furthest from development (Hamming 17 to 20), every model is correct.

99.42%

best accuracy, 95% CI 98.25 to 100%

0 of 36

model pairs significant after Holm

4.01 M

parameters in EfficientNetB0

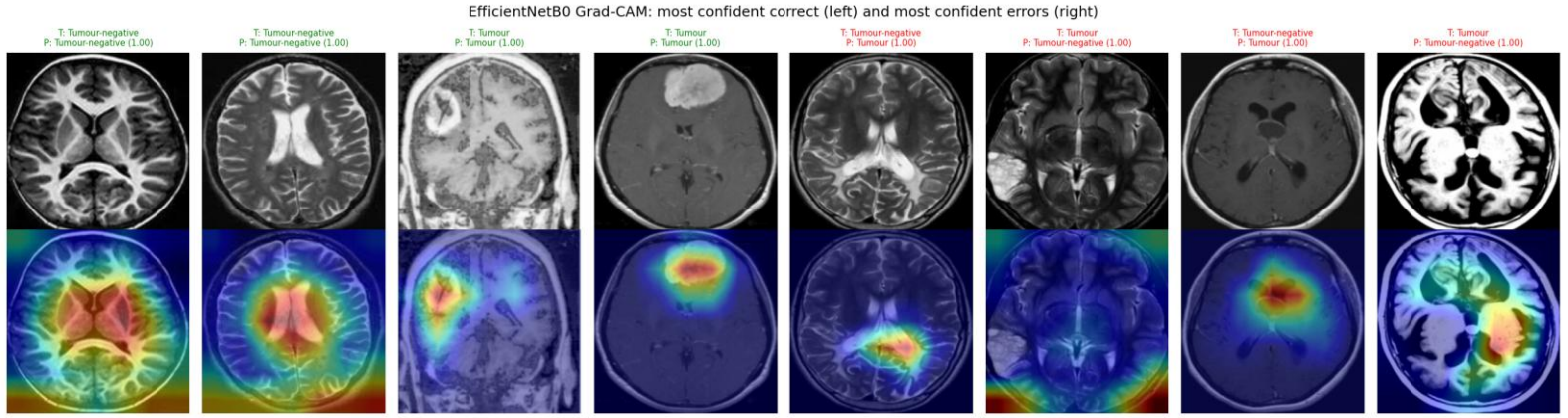
Subtype (45 test images)	Accuracy [95% CI]	Macro F1	MCC
ResNet50, DenseNet121, InceptionV3, ViT-B/16, Swin-Base, CoAtNet-0	100 [100, 100]	100	1.0000
VGG16, EfficientNetB0, MobileNetV3-Large	97.78 [93.33, 100]	87.56	0.9515

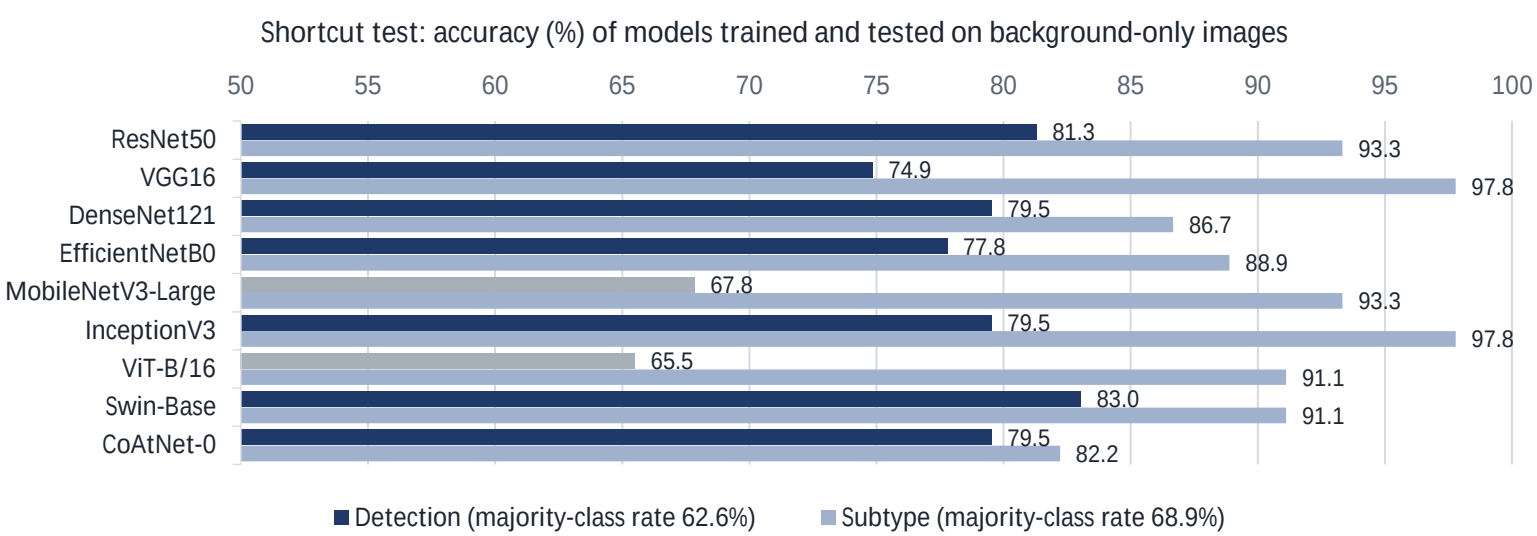
The single subtype error is a meningioma predicted as glioma; with one glioma among 45 test images, this test set cannot rank the models.

Calibration

- ECE is 0.0146 to 0.0326 (detection) and 0.0044 to 0.0371 (subtype).
- Temperature scaling lowered ECE only for ResNet50 and ViT-B/16 (detection) and CoAtNet-0 (subtype).
- Near-separable validation folds push the fitted temperature to its lower bound, $e^{-3} \approx 0.0498$, which sharpens the test probabilities.

Interpretability and Shortcut Audit





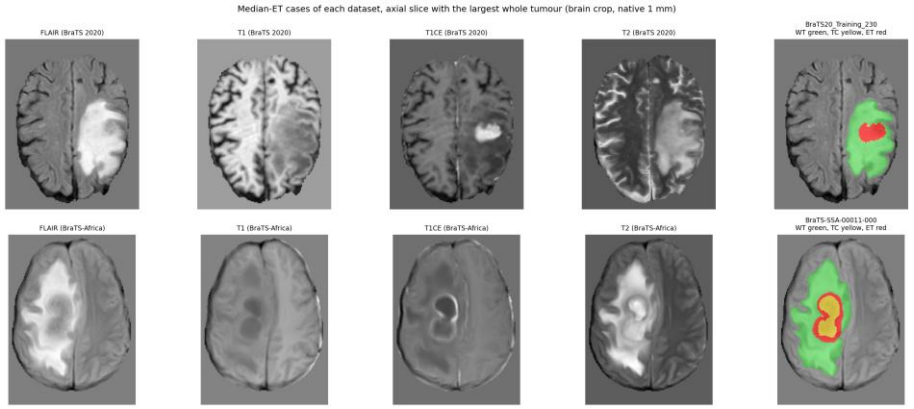
Where the models look

- Grad-CAM [85]: correct tumour predictions peak on the lesion; a missed temporal lesion peaks at the lower border
- ViT-B/16 attention rollout [86] puts 42.1% of its weight outside the head on tumour-negative images
- Occlusion sensitivity [87] outside the head: 0.7% to 19.4% (detection)

Masking and shortcut findings

- Head-only inputs (Otsu head mask [88]) keep 93.7% to 96.8% (detection) and 94.2% to 99.6% (subtype) accuracy
- Without the head: 44.4% to 53.3% (detection; InceptionV3 65.7%) and 11.6% to 33.8% (subtype)
- Retrained on background only, 7 of 9 detection and 9 of 9 subtype backbones beat the majority-class rate (binomial test, Holm) [44]; grey bars are not significant
- Borders and framing carry label information, so high accuracy on these collections does not prove that pathology was learned.

Data, Networks and Region-Adaptive Fusion



- BraTS 2020 [2, 89]: 369 cases (293 HGG, 76 LGG); split 257 / 56 / 56 by grade
- BraTS-Africa [13]: 60 external cases, never used for training or selection
- Nested regions: WT (all labels), TC (core and ET), ET; 27 of 76 LGG cases have no ET
- Brain crop and foreground z-score at native 1 mm; 128^3 patches, one third on tumour; rotation, scaling, mirroring, noise, brightness, contrast and gamma
- Gaussian 128^3 sliding windows with eight-flip TTA; ET size filter tuned on validation

Network	Type	Design (MONAI 1.6.1)	Params (M)	Epochs
nnU-Net [10]	CNN	Residual DynUNet, 6 stages, 32 to 320 ch.	31.42	60
SegResNet [11]	CNN	SegResNetDS, 32 filters, blocks 1-2-2-4	19.92	60
Swin UNETR-v2 [12, 37]	Transformer	Swin encoder, feature size 48	72.77	30

Region-adaptive hybrid ensemble

- For each region, a convex weight vector over the three networks (0.1 grid) and a threshold (0.3 to 0.7) maximise validation Dice
- Selected: WT 0.4 nnU-Net + 0.6 SegResNet (threshold 0.4); TC 0.4 / 0.5 / 0.1 Swin (0.3); ET 0.5 / 0.5 (0.4)
- Ablations: each network, mean of all, mean of the CNNs, the weights at threshold 0.5
- Soft Dice plus cross-entropy with deep supervision; 528 minutes of training on two Tesla T4 GPUs

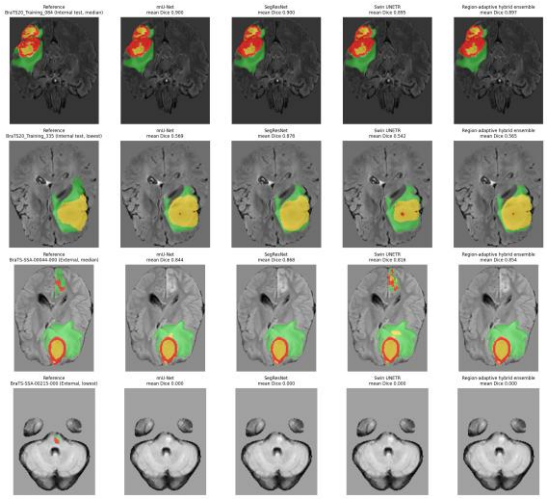
Internal and External Results

Model	Internal mean Dice [95% CI]	WT	TC	ET	External mean Dice [95% CI]
nnU-Net	0.8404 [0.8008, 0.8760]	0.9139	0.8493	0.7580	0.7765 [0.7224, 0.8203]
SegResNet	0.8717 [0.8389, 0.8987]	0.9264	0.8791	0.8095	0.7967 [0.7417, 0.8400]
Swin UNETR-v2	0.8360 [0.7923, 0.8742]	0.9162	0.8427	0.7490	0.7888 [0.7355, 0.8325]
CNN mean	0.8512 [0.8150, 0.8838]	0.9225	0.8630	0.7680	0.8005 [0.7484, 0.8419]
Mean of all three	0.8494 [0.8115, 0.8834]	0.9213	0.8601	0.7669	0.7971 [0.7456, 0.8400]
Proposed ensemble	0.8530 [0.8204, 0.8836]	0.9256	0.8639	0.7694	0.7966 [0.7430, 0.8391]

56 internal BraTS 2020 and 60 external BraTS-Africa test cases, eight-flip TTA; bootstrap CIs [82].

Statistical comparison

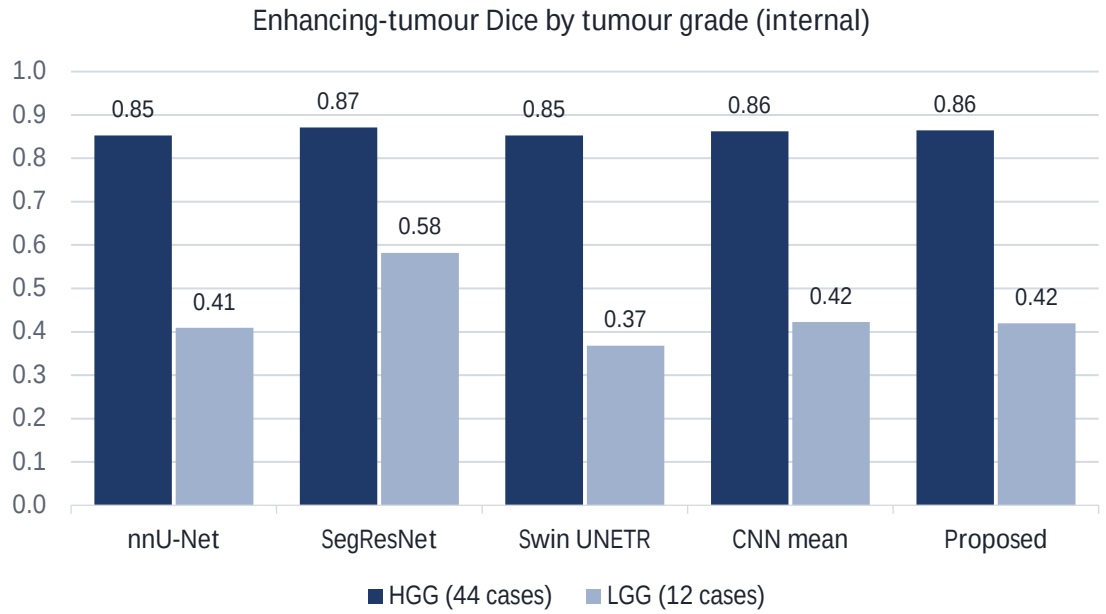
- Wilcoxon tests with Holm correction [84, 90]: the ensemble beats nnU-Net ($p = 0.0001$) and Swin UNETR ($p = 0.007$) internally
- SegResNet alone beats the ensemble internally ($p = 0.018$) and equals it externally
- Every model loses 0.047 to 0.075 Dice externally, mostly in TC

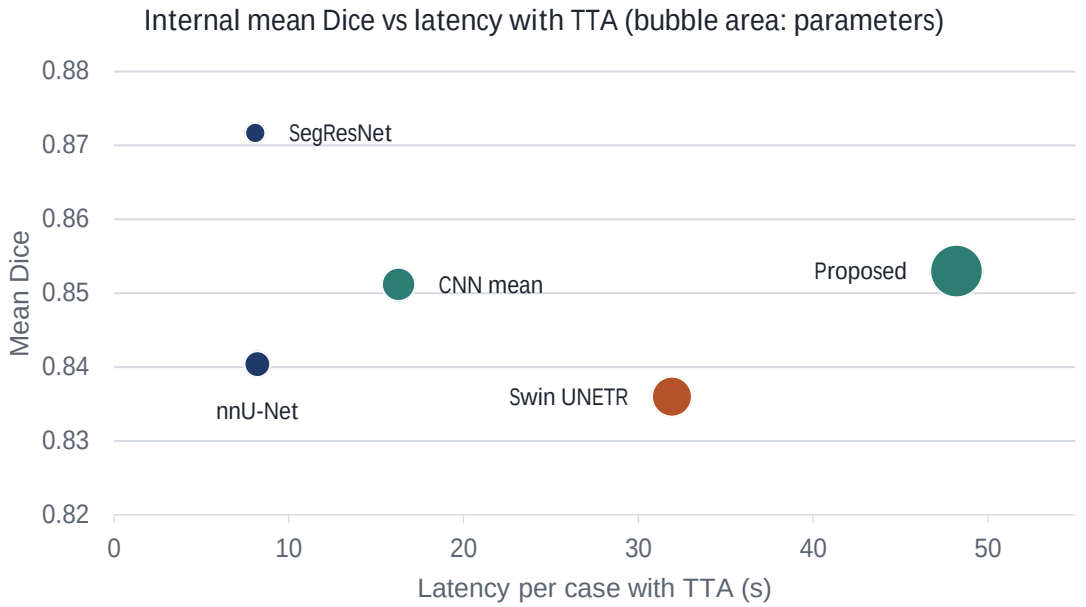


Qualitative comparison

- Rows: median and lowest internal case, median and lowest external case
- Lowest internal case: SegResNet 0.878, the other models 0.542 to 0.569
- Lowest external case: a small tumour that every model misses (Dice 0)
- Overlay: WT green, TC yellow, ET red

Tumour Grade, Quality Control and Efficiency





Disagreement flags failures

- 1 minus the mean pairwise Dice of the three networks' masks; failure: mean Dice below 0.80
- AUC 0.993 internal, 0.870 external [77]; Spearman −0.80 and −0.82 [91]
- TTA spread: AUC 0.716 and 0.791 [92–94]

Explanations, scored

- Seg-Grad-CAM [85, 95]: CNN maps 4 to 23 times enriched in the tumour; Swin UNETR 1.1 to 1.7 times
- Dropping FLAIR hurts WT, dropping T1CE hurts TC and ET, for every network
- Slice-wise Dice differs from per-patient 3D Dice [96]

SegResNet: the efficiency choice

- Smallest network (19.92 M) and fastest: 1.01 s per case, 8.08 s with TTA on a T4
- The ensemble needs all three networks: 124.11 M parameters, 48.20 s with TTA, six times SegResNet, for a lower internal Dice

Conclusion and Future Work

1. Cost decides in 2D. All nine backbones reach 97.08% to 99.42% detection accuracy with no significant difference; EfficientNetB0 matches the leaders with 4.01 M parameters and 0.77 GFLOPs.

2. Leakage and shortcuts are real. Deduplication removed 63.57% and 18.84% of the images, and background-only models still beat chance: public 2D collections carry non-anatomical label cues.

3. Fusion does not beat SegResNet in 3D. The region-adaptive ensemble (0.8530 internal, 0.7966 external) beats two of its networks but not SegResNet alone (0.8717, 0.7967), at six times its cost.

4. Disagreement is the useful product. Disagreement between the networks flags failed cases with AUC 0.993 internally and 0.870 externally; every model loses 0.047 to 0.075 Dice on BraTS-Africa, and LGG ET Dice is 0.37 to 0.58.

5. From benchmark to tool. NeuroScan AI deploys EfficientNetB0 and SegResNet in a secure, tested application; it is a research prototype, not a diagnostic device.

Future work

- Patient-level splits and external, multi-centre validation
- Grouping that keeps every subtype in the test set
- Standardise image borders, then repeat the shortcut test
- 3D multi-sequence detection and classification
- Several seeds and cross-validated fusion for the 3D study
- More external cohorts with raw clinical preprocessing
- Show member disagreement in the application; grade-aware ET post-processing
- Calibration on larger held-out sets; DICOM input and physical volumes in the application

Recommendation

- Deploy SegResNet (3D) and EfficientNetB0 (2D); use a second, different network to measure disagreement rather than to fuse
- Validate externally before trusting an internal ranking: it changed on BraTS-Africa

Outcome Achieved

Journal publication: two manuscripts prepared for *Engineering Applications of Artificial Intelligence* (Elsevier, Scopus indexed)

Manuscript 1

"Brain Tumour Magnetic Resonance Image Classification with Perceptual-Hash Deduplication: A Comparative Study Using Advanced Deep Learning Models"

P. Mital, V. Lawand, Biji C. L., M. A. Dar and G. P. Doss C.

Status: prepared for submission

Manuscript 2

"Region-Adaptive CNN-Transformer Ensembles for Brain Tumour Segmentation with External Validation and Disagreement-Based Quality Control"

P. Mital, V. Lawand and Biji C. L.

Status: prepared for submission

Others

- Open-source NeuroScan AI web application (TRL 4) with 64 backend and 39 frontend automated tests
- Research notebooks with all outputs: splits, per-run metrics, plots and the trained checkpoints
- Repository: https://github.com/parthmital/NeuroScan-AI

References (1/3)

[1] Q. T. Ostrom et al., “CBTRUS statistical report: Primary brain and other central nervous system tumors diagnosed in the United States in 2016–2020,” *Neuro-Oncology*, vol. 25, suppl. 4, pp. iv1–iv99, 2023.

[2] S. Bakas et al., “Advancing The Cancer Genome Atlas glioma MRI collections with expert segmentation labels and radiomic features,” *Scientific Data*, vol. 4, Art. no. 170117, 2017.

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